

US Banking Consumer Complaints Analysis

Risk & Resolution Analysis

Complete Project Handbook — Interview Preparation Guide

6,24,708

Complaints analyze kiye

3,193

Companies

36

Months ka data

4,255

FDIC banks matched

10

Pipeline stages

18+

Bugs pakde & fix kiye

Python 3.11

pandas

PostgreSQL 18

SQL Windows

rapidfuzz

SciPy

Tableau

Git

REST APIs

Is document mein kya hai

Ye handbook aapke poore project ka **A se Z** record hai — kya banaya, kaise banaya, kyun waise banaya, aur **interview mein har cheez ke baare mein kya bolna hai**. Har section mein ek hara box hai **"Interview mein aise bolna"** — wo aapki script hai.

- 1 Project ek nazar mein — problem, solution, impact
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ISE KAISE USE KAREIN

Interview se 2 din pehle: Section 1, 2, 13, 14, 15 padho — yahi 80% sawaal cover karte hain.

Interview se 1 ghanta pehle: Sirf hare boxes (Interview mein aise bolna) padh lo.

Technical round ke liye: Section 4–12 mein code aur decisions detail mein hain.

Project ek nazar mein

Ek line mein project kya hai

"Maine 6.24 lakh US banking complaints analyze karke ye prove kiya ki complaint *volume* se risk measure karna galat hai — aur size-adjust karne pe industry ki ranking poori tarah badal jaati hai."

Problem — jo galat ho raha tha

Har koi banks ko complaint **count** se judge karta hai. News headlines bhi yahi bolte hain — "Bank X ko sabse zyada complaints mile". Lekin ye **fundamentally misleading** hai, kyunki bada bank hone ki wajah se hi zyada customers hote hain, aur zyada customers ka matlab zyada complaints.

Ye bilkul aisa hai jaise kehna: "Mumbai mein sabse zyada crime hote hain, isliye Mumbai sabse khatarnak city hai" — bina population dekhe. Sahi metric hai **crime per lakh population**. Waise hi banking mein sahi metric hai **complaints per \$1B of assets**.

Solution — maine kya banaya

Ek complete end-to-end data pipeline: CFPB API se data pull kiya → pandas se clean kiya → FDIC ke bank asset data se enrich kiya (fuzzy matching se) → PostgreSQL mein load kiya → SQL window functions se analyze kiya → statistics se validate kiya → Tableau dashboard banaya.

Impact — kya nikla

Company	Volume rank	Size-adjusted rank	Kya matlab
Synchrony Financial	7	1	Volume pe normal, per dollar pe desh mein sabse kharab
Barclays Bank Delaware	13	2	11 position upar
SoFi Technologies	18	3	15 position upar
JPMorgan Chase	2	25 (of 28)	Volume pe 2nd worst dikhta hai, actually best mein se ek

SABSE POWERFUL LINE JO AAP BOL SAKTE HO

"Wells Fargo aur JPMorgan ko dekhiye — dono ko lagbhag same complaints mile, 41,051 aur 40,391. Volume chart pe ye twins lagte hain. Lekin JPMorgan ke paas **2.2 guna zyada assets** hain. Matlab per dollar Wells Fargo **2.2 guna zyada consumer friction** generate karta hai. Same data, bilkul ulta conclusion."

Numbers jo yaad rakhne hain

Metric	Value	Context
Total complaints analyzed	6,24,708	94 lakh mein se filter karke
Companies	3,193	112 ko score kiya (500+ complaints)
Time period	36 months	Jan 2023 – Dec 2025
National monetary relief rate	9.93%	Baseline for comparison
National untimely rate	2.32%	Baseline
Postgres load speed	36,700 rows/sec	COPY use karke
FDIC match rate	50.0%	Volume-weighted
Worst company (size-adj)	147.8 / \$1B	Synchrony — peer avg se 4.6×

Business problem — asli sawal kya tha

Primary question

YE POORA PROJECT ISI EK SAWAL KA JAWAB HAI

"Kaun si US financial companies sabse zyada consumer risk generate karti hain — aur company ke **size ko adjust karne ke baad**, kaun complaints ko sabse **kharab handle** karti hai (sabse kam monetary relief / sabse zyada late response)?"

Ye sawal kyun matter karta hai — 3 audiences

Kaun	Unke liye kyun useful
Regulators (CFPB, OCC)	Kis bank ki supervision badhani chahiye. Volume dekhoge toh hamesha bade banks hi dikhenge — chhote lekin risky banks chhoot jayenge.
Bank ki risk/compliance team	Apne peers ke against kahan khade hain. Agar aapka relief rate industry se bahut kam hai toh wo regulatory question ban sakta hai.
Consumers	Kis bank mein complaint karne se actually kuch hota hai. Relief rate 0% se 38.5% tak varies karta hai — 38 guna farak!

Supporting questions

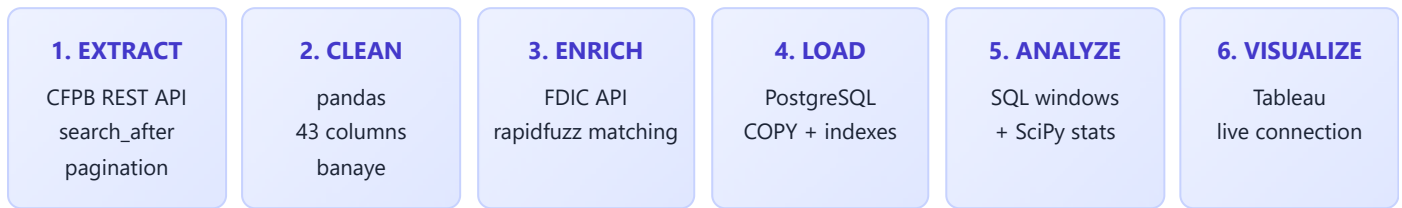
1. 2023 se products ke hisaab se complaint volume kaise badla, aur kaun se products accelerate kar rahe hain?
2. Kaun si companies raw volume pe theek lagti hain lekin size-adjust karne pe outlier ban jaati hain?
3. Kya complaint ka outcome (relief mila ya nahi) company se **statistically independent** hai?
4. Kaun se issues aur states mein risk sabse zyada concentrated hai?

INTERVIEW MEIN AISE BOLNA

"Project ka core insight ye hai ki **volume ek size proxy hai, risk signal nahi**. Maine CFPB ka data liya, usko FDIC ke bank asset data se join kiya, aur complaints per \$1B assets nikala. Isse ranking poori badal gayi. Jo companies volume chart pe invisible thi, wo top pe aa gayi. Aur maine ye sirf claim nahi kiya — chi-square test se statistically prove bhi kiya ki outcome company pe product se 1.8 guna zyada depend karta hai."

Architecture & tech stack

Pipeline ka flow



Har tool kyun choose kiya

Tool	Kis liye	Kyun yahi (interview mein ye poochha jaata hai)
Python 3.11	Poora pipeline	3.14 default tha machine pe, lekin pandas/scipy/psycopg2 ke stable wheels 3.11 pe hain. 3.14 pe source se compile karna padta. Reproducibility ke liye interpreter pin karna zaroori hai.
pandas	Cleaning, transformation	6 lakh rows ke liye perfect — memory mein aa jaata hai, aur vectorized operations fast hain.
PostgreSQL	Warehouse + analysis	Window functions (RANK, LAG), CTEs, proper types, indexes. Tableau ka native connector bhi hai.
rapidfuzz	Entity matching	C++ mein likha hai, fuzzywuzzy se 10-100× fast. 3,193 × 6,726 comparisons chahiye the.
SciPy	Chi-square test	Statistical validation — claim ko sirf number se nahi, test se back karna.
Tableau	Dashboard	Industry standard for BI. Live Postgres connection.
Git/GitHub	Version control	Har stage ka alag commit — reviewer decision history dekh sakta hai.

Repository structure

```

banking-complaints-project/
├── data/
│   ├── raw/           # CFPB output – gitignored, KABHI modify nahi kiya
│   ├── processed/     # cleaned outputs – gitignored
│   └── sample_1000.csv # sirf ye commit kiya, taaki schema dikhe
├── scripts/           # 01 se 09 tak numbered, re-runnable
├── sql/               # 6 files, har ek ek business question
├── notebooks/eda.ipynb # statistics
├── dashboard/         # Tableau guide + generated .twb
└── reports/           # cleaning_log, insights, figures
  
```

DESIGN PRINCIPLE JO POORE PROJECT MEIN FOLLOW KIYA

Har log file script se **GENERATE** hoti hai, haath se nahi likhi. Matlab `cleaning_log.md` mein jo numbers hain, wo `02_clean.py` chalane pe automatically likhe jaate hain. Isse ye kabhi purane nahi ho sakte. Agar data badla toh script dobara chalao, numbers apne aap update ho jayenge.

Ye interview mein bahut strong point hai — *"maine documentation ko code ka output banaya, manual step nahi"*.

Stage 0 & 1 — Setup aur Data Acquisition

STAGE 0 Project setup, Git, environment

Kya kiya?

Folder structure banaya, existing GitHub repo se connect kiya, Python virtual environment banaya, `.gitignore` configure kiya, aur `.env` se secrets alag kiye.

KYUN KIYA — YE DECISION IMPORTANT THA

Sabse important cheez: repo pehle se exist karta tha aur usme ek commit tha. Agar main `git init` karke naya history banata toh **duplicate/conflicting history** ban jaati. Isliye maine `fetch` karke existing `origin/main` ko track kiya — purani history preserve rahi.

Doosra: `.gitignore` mein raw data block kiya. 176 MB ka complaints.csv aur 405 MB ka narratives file kabhi GitHub pe nahi gaya. Sirf ek 1,000-row sample commit kiya taaki reviewer schema dekh sake.

Kaise kiya — technical detail

Python 3.11 pe venv banaya (3.14 default tha, lekin uspe libraries ke wheels nahi the). 111 packages install kiye aur `requirements-lock.txt` generate kiya — matlab exact versions record kiye taaki koi aur bhi bilkul same environment bana sake.

```
Verify kiya ki gitignore sach mein kaam kar raha hai:
$ git add --dry-run data/raw/complaints.csv
The following paths are ignored by .gitignore  BLOCKED

$ git add --dry-run data/sample_1000.csv
add 'data/sample_1000.csv'  ALLOWED
```

Result

Clean repo, reproducible environment, aur secrets safe. Poore project mein ek baar bhi password ya raw data GitHub pe nahi gaya — maine end mein `git log -p --all` se verify bhi kiya.

INTERVIEW MEIN AISE BOLNA

"Maine environment ko reproducible banane pe focus kiya. Interpreter version pin kiya, lock file banayi, aur gitignore ko sirf likha nahi — **test kiya**. Maine actually try kiya ki raw data commit ho paata hai ya nahi, aur confirm kiya ki block ho raha hai. Ye habit isliye important hai kyunki agar data ya password ek baar Git history mein chala gaya toh use hatana bahut painful hota hai."

STAGE 1 Data acquisition — CFPB API se 6.24 lakh rows

Kya kiya?

CFPB ke public API se filtered data pull kiya. Full bulk download 5–6 GB ka hai — wo deliberately avoid kiya. Pagination, retry logic, aur checkpoint system banaya.

KYUN KIYA — YE DECISION IMPORTANT THA

Yahan project ka sabse bada **scope decision** aaya. Date range mein **94,78,443** complaints the. Lekin unme se **82,30,677 (87%)** credit reporting ke the — Equifax, Experian, TransUnion ke against.

Ye credit bureaus **banks nahi hain**. Inke paas FDIC assets nahi hote. Matlab inka size-adjusted metric calculate hi nahi ho sakta. Aur 87% hone ki wajah se ye har chart, har product breakdown, har trend ko dabaa dete.

Isliye maine sirf **banking & lending products** rakhe — 10 categories jo koi bank ya lender actually bechta hai. Result: 94 lakh se 6.24 lakh rows.

Kaise kiya — technical detail

Pagination: API ka `from` offset parameter page 1 ke baad silently ignore ho jaata hai. Isliye Elasticsearch ka `search_after` cursor use karna pada — previous page ke last record ka sort value next request mein bhejte hain.

Retry: Exponential backoff with jitter — fail hone pe 2, 4, 8, 16 seconds wait, plus random jitter taaki saare retries ek saath na ho.

Checkpoint: Har page ke baad cursor file mein save karta hoon. Agar download beech mein ruk jaye toh dobara chalane pe wahin se resume hota hai, zero se nahi.

Narrative split: Complaint ka free-text 70% payload tha. Usko alag file mein `complaint_id` ke saath rakha — main table lean rahi (176 MB vs 580 MB).

```
# search_after cursor — previous page ke sort values se banta hai
if cursor:
    params["search_after"] = cursor

# Response se next cursor nikalna
cursor = "_".join(str(v) for v in hits[-1]["sort"])

# Exponential backoff + jitter
sleep_for = BACKOFF_BASE ** attempt + random.uniform(0, 1.5)
```

Result

6,24,727 rows — API ke reported total se bilkul exact match. 13/13 acceptance checks pass: zero duplicate IDs, zero unparsed dates, har product ka count scope table se match.

INTERVIEW MEIN AISE BOLNA

"Sabse important decision scope ka tha. 94 lakh rows available the, lekin 87% credit reporting complaints the jo credit bureaus ke against hain — wo banks hain hi nahi. Agar main unhe rakhta toh mera headline metric — complaints per \$1B assets — un companies ke liye calculate hi nahi ho paata, aur wo baaki sab data ko dabaa dete. Isliye maine banking products pe scope kiya. Ye analysis ko kamzor nahi, **zyada valid** banata hai — aur maine ye README mein clearly document kiya hai ki kya exclude kiya aur kyun."

"Technically, do cheezein interesting thi. Ek — API ka `frm` offset page 1 ke baad kaam nahi karta, toh `search_after` cursor use karna pada. Do — `format=json` bhejne se API export endpoint pe chala jaata hai jiski 1 lakh rows ki limit hai. Wo parameter hatate hi problem solve ho gayi. Ye dono maine documentation se nahi, **experiment karke** discover kiye."

Stage 2 — Data Cleaning (sabse important stage)

Ye stage sabse zyada showcase karne layak hai, kyunki yahan maine ek **aisi cheez pakdi jo analysis ko poori tarah galat kar deti**.

Problem #1 — Product rename ne fake trend bana diya

CFPB ne August 2023 mein apni product taxonomy badal di. Purane label band kar diye, naye shuru kar diye. Data mein ye aisa dikhta hai:

Month	"Credit card" (raw)	"Credit card or prepaid card" (raw)	Fix ke baad
2023-06	0	4,650	4,296
2023-07	0	5,252	4,891
2023-08	1,215	4,532	5,435
2023-09	4,918	0	4,918

AGAR YE FIX NAHI KARTE TOH KYA HOTA

Chart pe dikhta ki "Credit card complaints 0 se 4,918 ho gaye 3 mahine mein". Koi bhi analyst likh deta:

"Credit card complaints surged 300% in Q3 2023". Ye poori tarah galat hota — ye sirf label change tha, real surge nahi. Aur ye galti bilkul invisible hoti, kyunki chart perfectly normal dikhta.

Maine kaise fix kiya

Purane label "Credit card or prepaid card" ko `sub_product` ke basis pe wapas do naye labels mein **split** kiya. Ye **lossless** tha:

- 32,541 rows → "Credit card" (general-purpose + store credit card)
- 2,769 rows → "Prepaid card" (prepaid + gift + payroll + govt benefit)
- **Total 35,310 — bilkul exact, ek bhi row bachi nahi**

Result: August boundary pe max month-over-month swing 100%+ se girkar **11.1%** ho gaya.

Problem #2 — Company names merge ho rahe the (mera apna bug)

MAINE APNE HI CODE MEIN YE BUG PAKDA

Pehle version mein maine company name normalize karne ke liye `FINANCIAL` , `GROUP` , `BANCSHARES` jaise words strip kar diye the — soch ke ki ye "noise" hain. Result:

Merge ho gaya	Reality
<code>INDEPENDENT BANK CORP.</code> + <code>INDEPENDENT BANK GROUP, INC.</code>	Do bilkul alag banks! Ek Massachusetts ka (Rockland Trust), doosra Texas ka
<code>AMERICAN BANCSHARES MORTGAGE</code> + <code>AMERICAN FINANCIAL MORTGAGE</code>	Alag companies

Do alag banks ko merge karna matlab dono ka complaint count aur risk rate **corrupt** — aur koi symptom bhi nahi dikhta.

Fix — precision over recall

Ab sirf case, punctuation, aur **trailing** legal form (Inc, LLC, Corp) strip karta hoon. Beech ke words nahi chhedta. Result: 10 merges se ghatkar 4 merges, aur chaaron manually verify kiye.

YE TRADE-OFF KYUN SAHI HAI

Do alag banks ko merge karna (false positive) bahut zyada nuksaan karta hai bajaye ek hi bank ki do spellings ko merge na kar paane ke (false negative). Kyunki pehla case data ko silently corrupt karta hai, doosra case sirf ek company ranking se bahar rakhta hai.

Baaki cleaning decisions

Column	Null %	Decision	Reasoning
<code>tags</code>	83%	Fill 'No tag'	Ye missing data hai hi nahi. Field mein sirf 'Servicemember' ya 'Older American' aata hai. Null ka matlab "koi bhi nahi". Mode se impute karte toh 5 lakh fake veterans ban jaate!
<code>company_public_response</code>	55%	Fill 'No public response'	Company ne jaan-boojh ke public statement nahi di. Ye silence khud ek information hai.
<code>sub_issue</code>	8%	Fill 'Not specified'	CFPB taxonomy mein har issue ke sub-issues nahi hote. Null legitimate hai.
<code>state</code>	0.5%	Fill 'Unknown', row rakhi	State missing hone se complaint invalid nahi ho jaati. Company analysis ke liye wo row abhi bhi valid hai.

Missing values ki wajah se 0 rows drop ki.

Duplicates — kya hataya aur kya jaan-boojh ke nahi hataya

Test	Matches	Action	Kyun
<code>complaint_id</code> (source PK)	0	—	Source ki apni key unique hai
Business key + exact timestamp	19	HATAYE	Same company, same issue, same ZIP, same second — ye double submission hai
Business key + sirf date	9,568	NAHI HATAYE	Ye alag log hain jo same din same bank ke against complaint kar rahe hain — bilkul normal

YE DOOSRA WALA DECISION BAHUT IMPORTANT HAI

Agar main 9,568 rows hata deta toh ye **directional bias** create karta — sabse zyada complaints wali companies se sabse zyada rows delete hoti. Matlab jo companies sabse kharab hain, unki ranking automatically improve ho jaati. Dedup karke maine unhe bacha diya hota!

INTERVIEW MEIN AISE BOLNA

"Cleaning stage mein maine ek aisi cheez pakdi jo poore analysis ko galat kar deti. CFPB ne August 2023 mein product taxonomy rename ki thi. Raw data mein 'Credit card' complaints June aur July mein zero dikhte hain, phir September mein 4,918. Koi bhi ise dekh ke likh deta ki credit card complaints mein 300% surge aaya — jabki wo sirf ek rename tha."

"Maine purane label ko sub-product ke basis pe wapas split kiya, aur ye lossless tha — 32,541 plus 2,769 exactly 35,310 ban gaya. Iske baad month-over-month swing 11% pe aa gaya."

"Doosri cheez jo main highlight karunga wo ye hai ki maine **apne hi code mein bug pakda**. Mera pehla name-normalizer 'Independent Bank Corp' aur 'Independent Bank Group' ko merge kar raha tha — ye Massachusetts aur Texas ke do alag banks hain. Maine output review kiya, galti pakdi, aur logic ko conservative bana diya. Data cleaning mein **precision zyada important hai recall se**, kyunki galat merge silently data corrupt karta hai."

Stage 3 — Enrichment (FDIC fuzzy matching)

Ye project ka **differentiator** stage hai. Yahan external data laake wo metric banaya jo poore analysis ka base hai.

Matching problem

CORE CHALLENGE

CFPB **holding company** ka naam deta hai. FDIC **insured subsidiary** ka naam rakhta hai. Ye do alag naam hote hain:

CFPB kehta hai	FDIC mein hai
U.S. BANCORP	U.S. Bank National Association
TRUIST FINANCIAL CORPORATION	Truist Bank
SYNCHRONY FINANCIAL	Synchrony Bank

Do rules jo metric ko sahi rakhte hain

Rule 1 — Assets holding company level pe jodo

Ek CFPB company ke neeche kai FDIC charters ho sakte hain. Agar sirf sabse bade subsidiary ke assets use karo toh denominator kam ho jaata hai aur risk rate **artificially badh jaata hai**:

Holding company	Charters	Total assets	Sirf sabse bada	Kitna kam dikhta
Morgan Stanley	2	\$632.7B	\$391.3B	38.2%
Popular Inc	2	\$75.6B	\$60.6B	19.8%
Charles Schwab	3	\$280.8B	\$242.9B	13.5%

EK TRAP JO MAINE TIME PE PAKDA

683 institutions ke paas koi holding company hai hi nahi — unka `NAMEHCR` field blank hai. Agar main blank value pe group by karta toh **683 alag-alag independent banks ek hi fake \$822 billion ki entity ban jaate!** Aur jo bhi company usse match hoti, use wo absurd denominator mil jaata.

Fix: aise institutions ko unke apne `CERT` number se key kiya — har ek alag rehta hai.

Rule 2 — Unmatched companies ko drop mat karo

50% complaint volume aisi entities ka hai jinke FDIC assets hain hi nahi — **by design**:

Type	Complaints	Example
Credit bureaus	~57,000	Equifax, TransUnion, Experian
Credit unions (NCUA-insured)	~26,000	Navy Federal
Loan servicers	~50,000	MOHELA, Nelnet, Navient
Fintechs	~30,000	Chime, Block, SoFi ka non-bank part

Ye sab **volume, trend, aur resolution analysis mein rehte hain**. Sirf size-adjusted ranking se bahar hain. Aur maine "unmatched but high-volume" ki alag list publish ki — kyunki *ye khud ek finding hai* ki itna bada hissa prudential regulation ke bahar hai.

Fuzzy matching — har guard ek real bug se aaya

Ye section interview mein sabse zyada impress karta hai, kyunki ye dikhata hai ki maine output **review** kiya, blindly trust nahi kiya.

Jo galat match mila	Score	Kyun hua	Guard jo add kiya
Ocwen Financial → OWEN FINANCIAL CORP	96.6	Ek letter ka farak! Ocwen ek mortgage servicer hai, Owen ek \$0.4B ka chhota bank. Result: 7,923 complaints per \$1B — bilkul bakwas number	Kam se kam 1 distinctive token exactly match hona chahiye
Bank of America → reject ho gaya	100	Over-correction! Iske saare tokens (BANK, OF, AMERICA, NATIONAL, ASSOCIATION) generic hain, toh koi distinctive token tha hi nahi	Exact normalized string match ko hamesha accept karo
MECHANICS BANK → Farmers and Mechanics Federal	92+	Sirf ek token 'MECHANICS' share ho raha tha	Token-set Jaccard ≥ 0.6
Paramount GR Holdings → PARAMOUNT FINANCIAL GROUP	92+	Dono ka distinctive token set identical tha — lekin sirf ek token	Kam se kam 2 shared tokens chahiye
FIFTH THIRD FINANCIAL → PATHWARD FINANCIAL	—	Sahi candidate list mein tha, lekin galat wale ne zyada score kar liya	Token rule ko top-25 candidates pe filter ki tarah lagao, sirf top-1 pe check nahi

STRICTNESS KA COST AFFORDABLE KYUN THA

Maine data dekha: `exact_normalised` aur curated overrides milke **99% matched volume** carry karte hain. Fuzzy paths sirf 1% volume de rahe the — **lekin saare false positives wahi se aa rahe the**. Toh strict banana lagbhag free tha.

Final safety net

Ek **plausibility sweep** banaya: agar kisi matched company ka ratio 300 complaints per \$1B se zyada hai toh flag karo — koi real bank itna generate nahi kar sakta. Abhi ye khaali return karta hai. Aur main isko **report** karta hoon, auto-correct nahi — taaki future mein koi regression aaye toh dikhe, chhupe nahi.

INTERVIEW MEIN AISE BOLNA

"Enrichment mein maine CFPB companies ko FDIC ke 4,255 institutions se match kiya rapidfuzz se. Challenge ye tha ki CFPB holding company ka naam deta hai aur FDIC subsidiary ka."

"Sabse important baat — **maine har guard ko design karke nahi, output review karke add kiya**. Mera pehla matcher 'Ocwen Financial' ko 'Owen Financial Corp' se match kar raha tha — ek letter ka farak, 96.6 score. Ocwen mortgage servicer hai, Owen \$0.4 billion ka chhota bank. Isse 7,923 complaints per billion ka number aa raha tha, jo obviously galat hai."

"Maine fix kiya toh over-correction ho gaya — Bank of America reject hone laga, kyunki uske naam mein koi distinctive word hi nahi hai. Toh doosra rule add karna pada. Total 5 aise guards bane, har ek kisi real false positive ki response mein."

"Aur maine ek plausibility check chhoda hai jo 300 per billion se zyada ratio ko flag karta hai. Abhi wo khaali hai, lekin agar future mein koi regression aayi toh turant dikh jayegi. **Main auto-correct nahi karta — report karta hoon**, kyunki silent fix se problem chhup jaati hai."

Stage 4 — PostgreSQL + SQL Analysis

STAGE 4 Typed warehouse, bulk load, indexes, 6 analysis queries

Kya kiya?

Database banaya, typed tables define kiye, COPY se bulk load kiya, 9 indexes banaye, integrity checks chalaye, aur 6 SQL analysis files likhi.

KYUN KIYA — YE DECISION IMPORTANT THA

Types declare kiye, infer nahi kiye. Sab kuch TEXT rakhna kaam kar jaata lekin galat hota — date arithmetic, BETWEEN filters, AVG(), boolean predicates sabko real types chahiye. Aur Tableau column types dekh ke decide karta hai ki kya measure hai aur kya dimension.

Kaise kiya — technical detail

COPY vs INSERT: 6.24 lakh rows row-by-row INSERT se minutes lagte. PostgreSQL ka COPY command poori CSV ko ek hi statement mein stream kar deta hai. psycopg2 mein `copy_expert` se ye expose hota hai — file Python mein parse hoti hi nahi.

Indexes: company, product, state, date pe single-column, plus do composite indexes jo actual query shapes se match karte hain: `(company, is_resolved, in_trend_window)` aur `(product, year_month)`.

```
# Fast bulk Load - 36,700 rows/second
copy_sql = f"COPY {SCHEMA}.{name} ({cols}) FROM STDIN "
          f"WITH (FORMAT csv, HEADER true, NULL '')"

with csv_path.open("r", encoding="utf-8") as fh:
    cur.copy_expert(copy_sql, fh) # file Python mein parse hi nahi hoti
```

Result

6,34,961 rows load hue 17 seconds mein — 36,700 rows/second. 7/7 integrity checks pass: row count, unique IDs, zip3 leading zeros, DATE/BOOLEAN types (information_schema se verify), date bounds, referential coverage.

INTERVIEW MEIN AISE BOLNA

"Load ke liye maine `copy_expert` use kiya, pandas ka `to_sql` nahi. `to_sql` batched INSERTs karta hai jo 6 lakh rows ke liye minutes leta hai aur WAL flood karta hai. COPY ne 17 second mein kar diya — 36,700 rows per second."

"Ek subtle cheez jo maine catch ki: `zip3` column ko TEXT rakha, numeric nahi. 46,528 rows mein leading zero hai — jaise 007 Puerto Rico ka hai. Numeric padho toh 007 ban jaata hai 7, aur koi bhi ZIP reference data se join toot jaata hai. CSV theek tha — damage read pe hota hai. Isliye har read mein dtype force kiya aur ek integrity check likha jo verify karta hai ki leading zeros bache hain."

SQL files — har ek ek business question

File	Question	Technique
01_company_complaint_ranking	Kaun si companies sabse zyada complaints generate karti hain?	RANK(), DENSE_RANK(), NTILE(), running cumulative share
02_monthly_trend_by_product	Month-over-month trend kya hai?	LAG(1), LAG(12) for YoY, LEAD(), 3-month moving average
03_resolution_rates	Kaun complaints sabse kharab resolve karta hai?	FILTER aggregates, PERCENT_RANK(), peer median CTE
04_size_adjusted_risk	HEADLINE — size adjust karke kaun riskiest hai?	FDIC join, dual RANK(), rank gap calculation
05_top_issues_by_product	Har product mein top issues kya hain?	ROW_NUMBER() PARTITION BY
06_top_issues_by_state	Geographic distribution kya hai?	ROW_NUMBER() PARTITION BY, vs-national comparison

SQL KA WO HISSA JO INTERVIEWER KO DIKHANA CHAHIYE

Query 04 mein maine **rank gap** nikala — volume rank minus size-adjusted rank. Ye number hi asli finding hai. Positive gap ka matlab: company volume pe safe dikhti hai lekin actually nahi hai. Ye ek single column mein poora insight capture kar leta hai.

```
CASE WHEN volume_rank - size_adjusted_rank >= 5
  THEN 'UNDERSTATED by volume'
  WHEN volume_rank - size_adjusted_rank <= -5
  THEN 'overstated by volume'
  ELSE 'consistent'
END AS volume_vs_size_verdict
```

Stage 5 — Statistics (chi-square + effect size)

Sabse important statistical concept jo is project mein hai

YE BAAT INTERVIEW MEIN AAPKO ALAG DIKHAYEGI

Jab sample size 6 lakh ho, toh **p-value bekaar ho jaata hai**. Chi-square test kisi bhi association pe $p < 0.001$ de dega — chahe wo association kitni bhi choti ho. Matlab "statistically significant" hona guaranteed hai, aur isliye wo koi information nahi deta.

Isliye maine har test ke saath **Cramér's V** report kiya — ye **effect size** hai, jo batata hai association kitni **strong** hai, sirf ye nahi ki exist karti hai ya nahi.

Test results

Test	Chi-square	df	p-value	Cramér's V	Interpretation
Outcome × Product	21,575	7	$< 1e-300$	0.186	Weak
Outcome × Company	65,031	111	$< 1e-300$	0.341	Strong

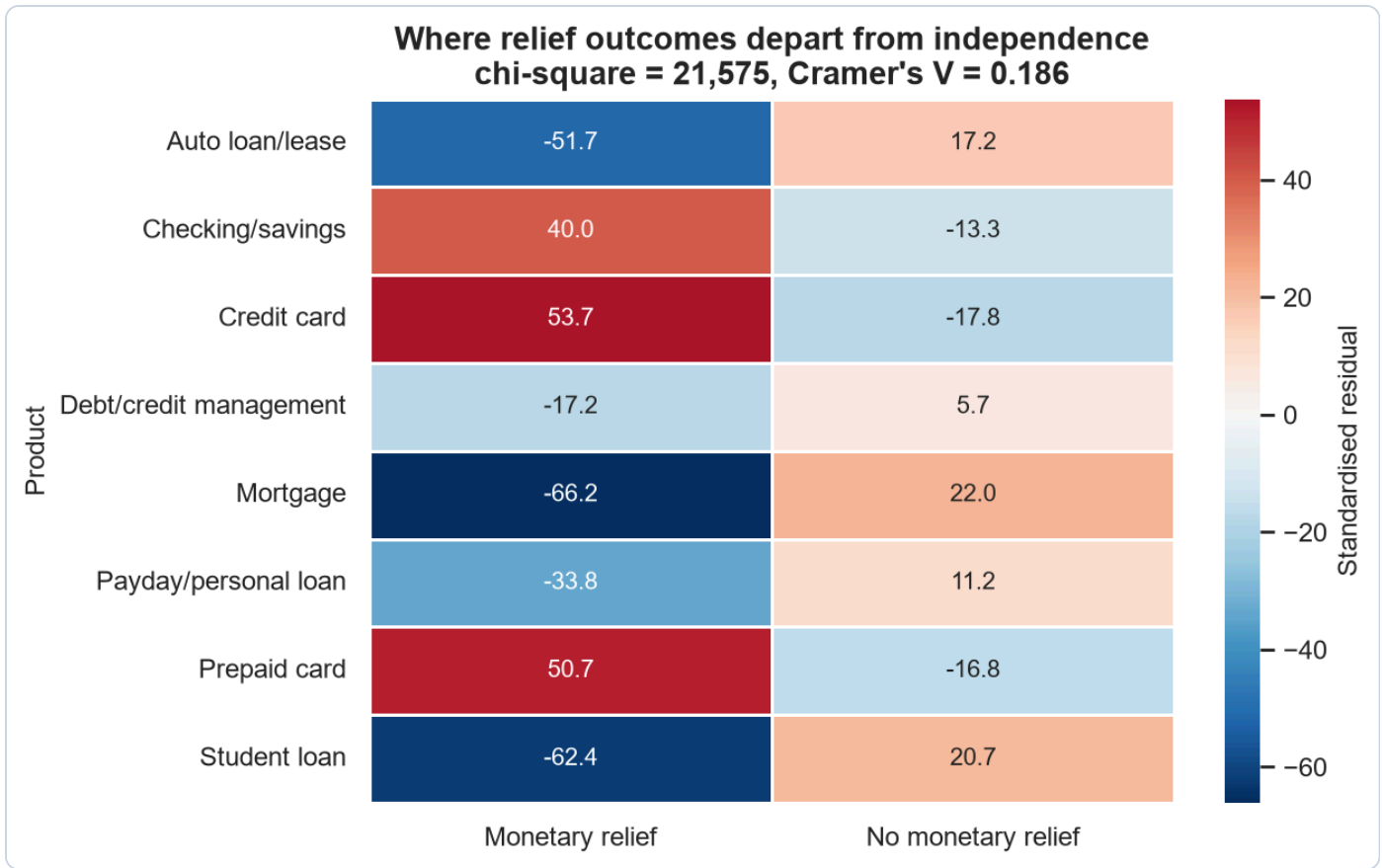
Company ka association product se 1.8 guna strong hai.

Matlab: aap kis company ke against complaint kar rahe ho, ye zyada matter karta hai bajaye iske ki aap kis cheez ke baare mein complaint kar rahe ho.

Standardized residuals — kaun se cells result drive kar rahe hain

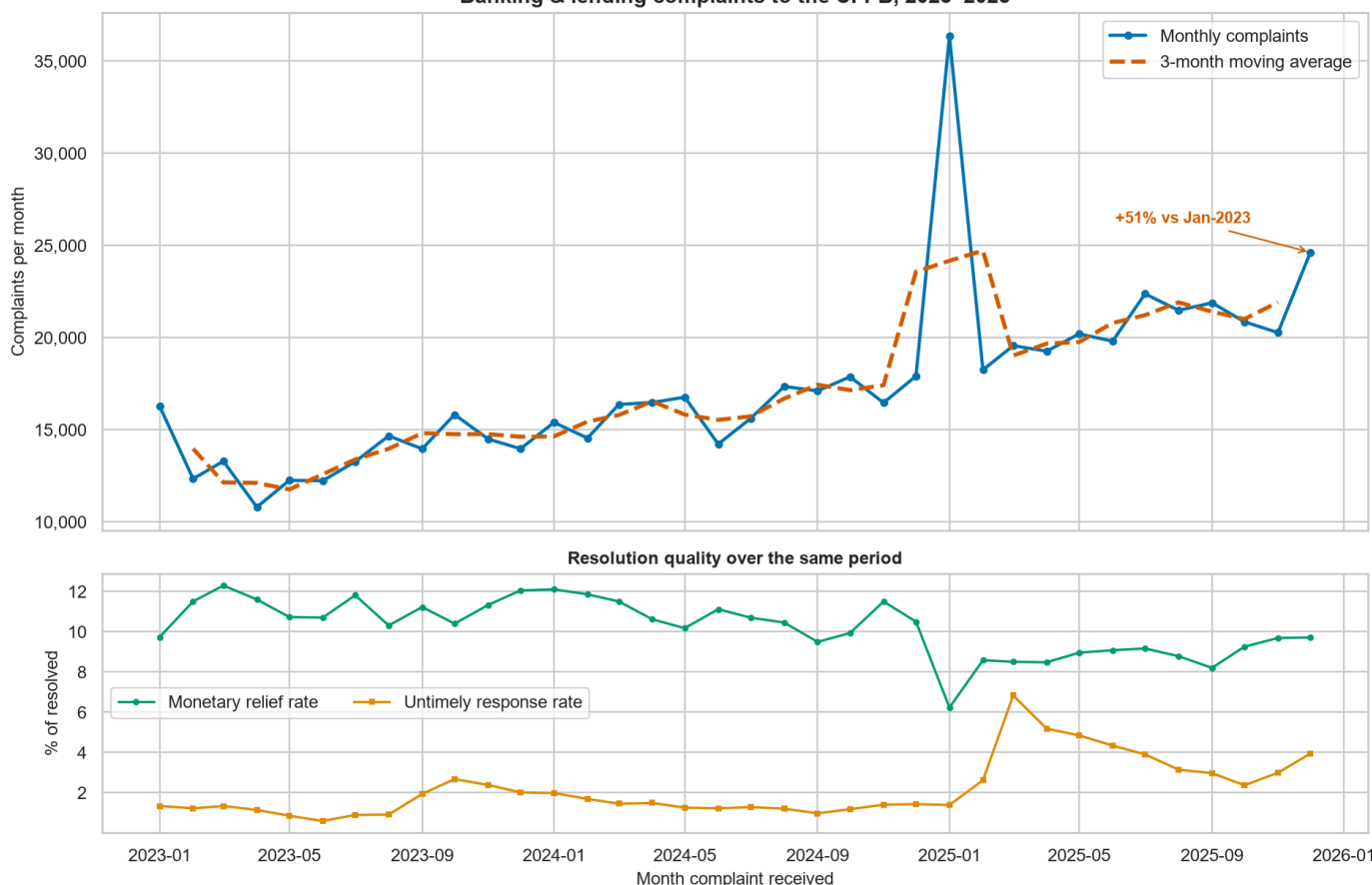
Chi-square ek single number deta hai poore table ke liye. Lekin ye nahi batata ki *kahan* problem hai. Standardized residuals se pata chalta hai — ± 2 se bahar wale cells hi asli kaam kar rahe hain:

Product	Residual	Matlab
Mortgage	-66.2	Expected se bahut kam relief milta hai
Student loan	-62.4	Expected se bahut kam relief
Auto loan	-51.7	Expected se kam relief
Credit card	+53.7	Expected se zyada relief
Prepaid card	+50.7	Expected se zyada relief



Anomaly jo maine chart dekh ke pakdi

Banking & lending complaints to the CFPB, 2023–2025



JANUARY 2025 — 36,342 COMPLAINTS (NORMAL 18,000)

Trend chart banate hi ek spike dikha jo turant wapas normal ho gaya. Spike jo revert ho jaye wo aam taur pe organic demand nahi hoti. Maine investigate kiya:

- **18,441 extra complaints**
- **67% sirf do companies se** — Navy Federal (484 → 7,725, **16 guna**) aur Capital One (840 → 5,876)
- 15–18 January pe concentrated, 17 tarikh ko akele 4,978
- Dominant issue: overdraft/NSF fees

Ye ek **coordinated filing campaign** hai — aam taur pe kisi enforcement action ke baad hoti hai. Service quality kharab hone ka signal nahi hai.

Maine kya kiya — aur kyun delete nahi kiya

Complaints **real** hain, isliye maine unhe delete nahi kiya. Balki flag kiya, aur ek **sensitivity check** chalaya: mahine ko hata ke ranking dobara calculate ki.

Result: 28 mein se 28 companies ka rank same raha.

Matlab ranking is event se robust hai — aur ab ye **proven** hai, assumed nahi.

INTERVIEW MEIN AISE BOLNA

"Statistics mein maine ek cheez pe dhyan diya jo bahut log miss karte hain. Mera sample 6 lakh ka hai — is size pe chi-square har association pe p less than 0.001 dega. Significance guaranteed hai, isliye wo useless hai. Isliye maine **Cramér's V** report kiya, jo effect size batata hai."

"Result ye tha ki company ka effect 0.34 tha aur product ka 0.19 — matlab company 1.8 guna zyada matter karta hai. Ye ek business-relevant statement hai: aap kis bank ko complaint karte ho ye zyada important hai bajaye iske ki complaint kis baare mein hai."

"Aur maine trend chart banate waqt ek anomaly pakdi — January 2025 mein spike tha jo turant revert ho gaya. Investigate kiya toh pata chala 67% spike sirf do companies se aa raha tha, 4 din mein concentrated, overdraft issues pe. Ye coordinated filing campaign thi. Maine data delete nahi kiya kyunki complaints real hain — balki flag kiya aur sensitivity check chalaya. 28 mein se 28 companies ka rank same raha, toh ranking robust hai. **Maine assume nahi kiya, verify kiya.**"

Stage 6 & 7 — Risk Score aur Tableau

Stage 6 — Do scores banaye, ek nahi

EK SINGLE COMPOSITE SCORE KYUN GALAT HOTA

Reason 1: `complaints_per_1b_assets` balance sheet size se divide karta hai, customer count se nahi. Ek credit card company lakhs customers serve karti hai chhote balance sheet pe; ek universal bank ke paas mortgages aur commercial loans hote hain jo assets badhate hain bina retail customers badhaye. Toh card issuers **business model ki wajah se** upar aate hain, conduct ki wajah se nahi.

Reason 2: Ye metric 50% complaint volume ke liye **undefined** hai (non-banks). Single composite chupchap aadha market drop kar deta.

Score	Kiske liye	Kya measure karta hai
<code>conduct_risk_score</code>	Har company (112)	Sirf resolution behaviour. Business model se distort nahi hota. Conduct claims ke liye yahi use karna
<code>exposure_risk_score</code>	FDIC-matched (28)	Isme complaint frequency per \$1B bhi add hoti hai

Score formula — percentile based

Component	Weight	Kya capture karta hai
<code>pct_no_relief</code>	40%	Consumer ko kuch bhi nahi mila — na paisa, na non-monetary relief
<code>untimely_rate</code>	35%	Deadline miss ki — hard failure, poori tarah company ke control mein
<code>disputes_facts_rate</code>	15%	Company consumer ki baat ko formally challenge karti hai
<code>avg_days_to_company</code>	10%	Routing speed — weakest signal, isliye kam weight

EK COMPONENT MAINE TEST KARKE HATA DIYA

Pehle version mein `explanation_only_rate` bhi tha 20% weight pe. Maine check kiya toh wo `pct_no_relief` se **Spearman ρ = 0.954** pe correlate kar raha tha, aur average sirf **1.0 percentage point** ka farak tha.

Matlab ye **ek hi cheez do baar count ho rahi thi** — explanation ke saath close karna hi toh no relief dena hai. Dono milke 55% weight le rahe the, aur `untimely_rate` ko dabaa rahe the.

Hataane ke baad ranking sahi hui: "Servicer under contract with Federal Student Aid" — jo 100% no relief **aur** 100% untimely hai — top pe aaya, jo bilkul sahi hai.

Stage 7 — Tableau handoff

4 PostgreSQL views banaye taaki Tableau clean, pre-shaped data padhe:

View	Rows	Kis chart ke liye
v_company_summary	112	Size-adjusted ranking + resolution scatter
v_monthly_trend	281	Trend line (pre-aggregated, fast)
v_state_summary	51	Map / state chart
v_complaints_enriched	6,24,195	Detail aur free-form filtering

VIEWS KYUN BANAYE, TABLEAU CALCULATED FIELDS KYUN NAHI

View ka matlab hai ki **business logic Tableau mein nahi rehta**. 500-complaint floor, resolved-only denominator, trend window bound, FDIC join — sab SQL mein hain jahan wo versioned aur reviewable hain. Agar "risk" ki definition badalti hai toh view badlo, saare worksheets automatically follow karenge. Kisi ko yaad nahi rakhna padega ki kaun se calculated field mein rule chhupa tha.

Bonus — poora Tableau workbook code se generate kiya

`.twb` file plain XML hoti hai. Toh maine ek script likhi jo poora workbook generate karti hai — 4 worksheets, dashboard, sab pre-built. File open karo, sab ready.

Isme 4 round debugging lagi kyunki Tableau ka XML schema bahut strict hai. Jo errors mile:

Error	Fix
missing required attribute 'source-build'	Workbook tag mein build version add ki
missing elements in content model ...aggregation)	Har <code><view></code> ka aakhri child <code><aggregation></code> hona chahiye
value 'Map' not in enumeration	Mark class 'Map' valid nahi hai — Bar use kiya
no declaration found for element 'natural-sort'	Tableau khud ye element likhta hai apni saved files mein, lekin apna hi loader use reject karta hai! Computed <code><sort></code> use kiya

INTERVIEW MEIN AISE BOLNA (STAGE 6 & 7)

"Risk score mein maine ek deliberate design choice ki — **do scores banaye, ek nahi**. Kyunki size-adjusted metric business model se biased hai: card companies chhote balance sheet pe lakhs customers serve karti hain, toh wo automatically upar aati hain. Aur wo metric 50% volume ke liye defined hi nahi hai. Isliye ek pure conduct score banaya jo sabke liye valid hai."

"Score banate waqt maine components ko **test kiya**, assume nahi kiya. Do components 0.954 pe correlate kar rahe the — matlab ek hi cheez do baar count ho rahi thi, 55% weight pe. Ek hata diya aur rebalance kiya. Iske baad worst performer sahi jagah aaya."

"Tableau ke liye maine views banaye taaki business logic SQL mein rahe, workbook mein nahi. Aur ek step aage jaake maine poora Tableau workbook **code se generate** kiya — .twb XML hi toh hai. Isse dashboard reproducible ban gaya: script chalao, wahi workbook milega, bajaye iske ki jo us din click kiya tha."

Bugs jo maine pakde — aapka biggest differentiator

YE SECTION KYUN SABSE IMPORTANT HAI

Har candidate keh sakta hai "maine pipeline banaya". Bahut kam log ye keh sakte hain **"maine apne hi kaam mein ye galtiyan pakdi aur fix ki"**. Ye seniority ka sabse strong signal hai — kyunki iska matlab hai ki aapne output **verify** kiya, blindly trust nahi kiya.

Agar interviewer poochhe "sabse mushkil part kya tha?" — **yahan se answer do**.

Category A — Data bugs (analysis galat kar dete)

#	Bug	Impact agar na pakadte
1	Product taxonomy rename — Aug 2023 mein CFPB ne labels badle	"Credit card complaints 300% surge" — bilkul jhoota claim, aur chart normal dikhta
2	Ocwen → Owen false match — ek letter ka farak, 96.6 score	7,923 complaints/\$1B — nonsense number top pe dikhta
3	Independent Bank Corp + Group merge — do alag banks	Dono ka count aur risk rate corrupt, bina kisi symptom ke
4	rank_gap alag populations compare kar raha tha — volume rank 112 mein se, size rank 28 mein se	Har gap inflated. SoFi "32 → 3 (+29)" dikh raha tha, sahi hai "18 → 3 (+15)"
5	zip3 leading zeros — 007 (Puerto Rico) → 7	46,528 rows ka ZIP join toot jaata
6	Jan 2025 mass filing event	Coordinated campaign ko real service degradation samajh lete
7	2026 ka one-day stub — window 1 Jan ko khatam	Trend chart mein 100% collapse dikhta, jo actually missing data hai

Category B — Code bugs (silently galat behave karte)

#	Bug	Kyun subtle tha
8	'None' sentinel — column ko 'None' string se fill kiya	pandas ka default <code>na_values</code> mein 'None' shaamil hai! CSV likh ke padho toh saare nulls wapas aa jaate hain. Fill khud ko undo kar deta hai
9	Score components collinear — $\rho = 0.954$	Ek hi cheez 55% weight pe do baar count ho rahi thi
10	.gitignore inline comments	<code>hyperd.log # comment</code> — gitignore inline comments support nahi karta, poori line pattern ban jaati hai, rule dead ho jaata hai
11	<code>hash()</code> non-determinism	Python string hashes har process mein randomize hote hain — generated file har baar alag banti
12	Generator manual edits mita raha tha	Tableau mein save karne ke baad script chalao toh saara kaam gayab

Category C — Platform gotchas (documentation mein nahi milte)

#	Gotcha	Kaise pata chala
13	CFPB <code>format=json</code> export endpoint pe bhej deta hai — 1 lakh rows ki cap	Experiment karke — parameter hata ke test kiya
14	<code>frm</code> offset page 1 ke baad silently ignore	Page 2 ne same rows return kiye
15	FDIC search ranking reliable nahi — 'TRUIST' pe JPMorgan first aata hai	Isliye saara data local pull karke match kiya
16	<code>CREATE DATABASE</code> psycopg2 ke context manager mein nahi chalta	Transaction block error
17	<code>PERCENTILE_CONT</code> window function ki tarah use nahi ho sakta	Postgres error — alag CTE mein nikala
18	Tableau ka <code>natural-sort</code> — Tableau khud likhta hai lekin apna loader reject karta hai	Workbook load error

"SABSE MUSHKIL PART KYA THA?" — KA JAWAB

"Sabse mushkil technically nahi, **discipline ka tha** — apne hi output pe bharosa na karna."

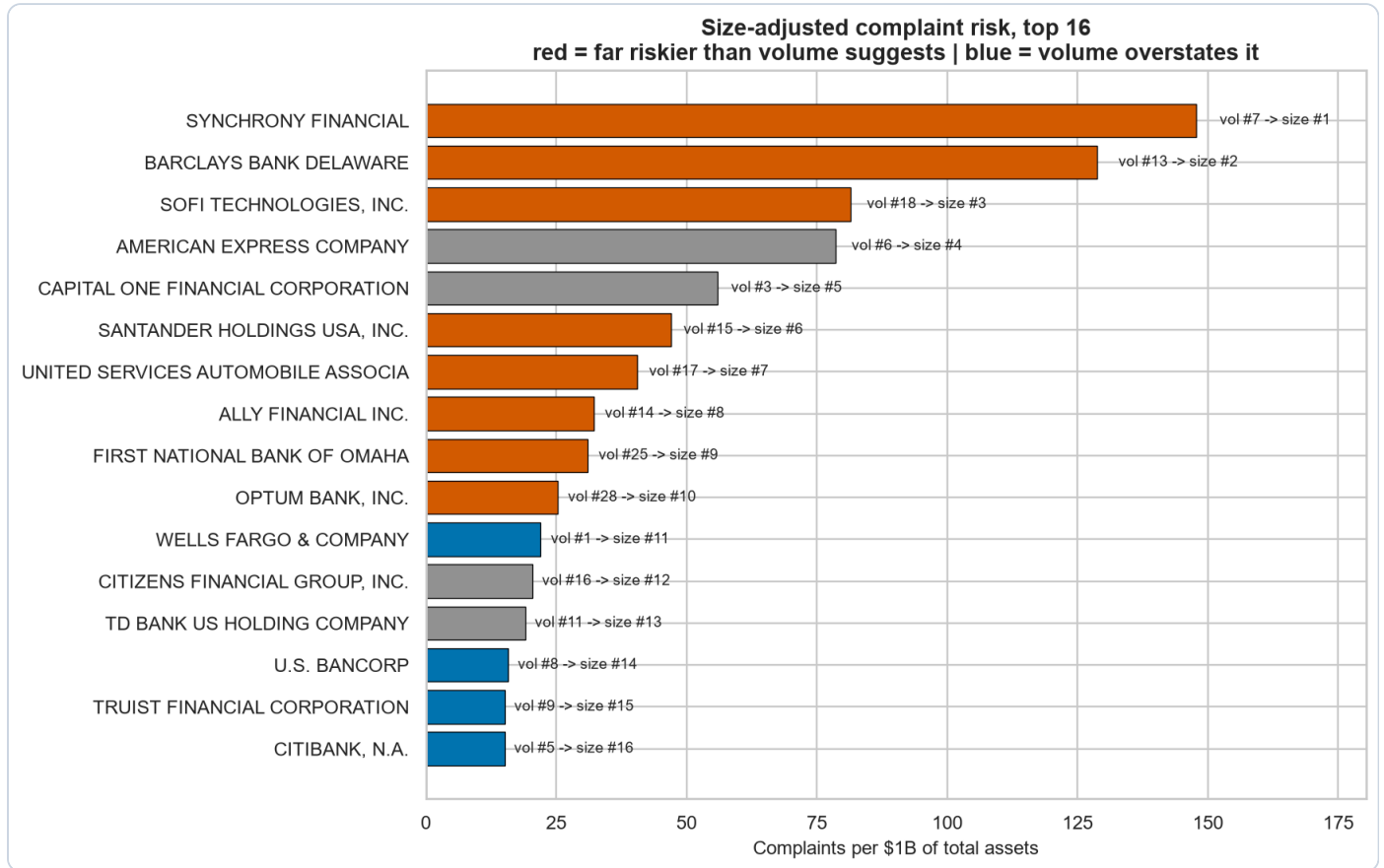
"Ek example deta hoon. Fuzzy matching ke baad maine results ki list dekhi. Ek company thi Ocwen Financial jiska ratio 7,923 complaints per billion tha. Ab main chahta toh isse accept kar leta — score 96.6 tha, high confidence. Lekin number obviously absurd tha. Check kiya toh pata chala wo 'Owen Financial Corp' se match hui thi — ek letter ka farak, aur wo ek \$0.4 billion ka chhota bank hai jabki Ocwen mortgage servicer hai."

"Isko fix karne mein **5 round lage**, kyunki har fix ne naya problem banaya. Pehle fix ne Bank of America reject kar diya. Uska fix karne pe Mechanics Bank galat match hone laga. Har baar mujhe output dobara review karna pada."

"Isse main ye seekha ki fuzzy matching mein **failure mode chunna padta hai**. Maine chuna ki galat denominator lagane se accha hai ki denominator na lage — kyunki galat match confidently galat answer deta hai, aur no match sirf company ko ek ranking se bahar rakhta hai. Aur maine ek plausibility sweep chhoda jo abhi khaali hai, taaki future regression dikhe."

Final Results — 4 Business Insights

Insight 1 — Volume risk signal nahi hai



Size-adjust karne pe industry ki ranking badal jaati hai. Synchrony 7th se 1st, JPMorgan 2nd se 25th (28 mein se). Synchrony ka rate **147.8 per \$1B** hai — peer average 32.3 se **4.6 guna**, aur JPMorgan se **14.7 guna**.

RECOMMENDATION

Risk/supervisory teams ke liye: complaint counts pe ranking band karo. Complaints per \$1B assets ko primary screening metric banao. Jin institutions ka size-adjusted rank unke volume rank se 10+ positions upar hai (SoFi, Barclays, Santander), unko review karo — volume monitoring se ye kabhi trigger nahi honge.

Insight 2 — Student loan servicing systemic failure hai

Company	Complaints	Untimely rate	vs national (2.32%)	Relief
"Servicer under contract with Federal Student Aid"	1,667	100.0%	43×	0.00%
EdFinancial Services	4,014	43.3%	19×	0.00%
MOHELA	17,225	27.2%	12×	1.57%

Ek entity ne **1,667 mein se har ek** complaint ka jawab late diya. Product level pe: student loan complaints credit card se **59 guna zyada** late jawab paate hain aur **17 guna kam** relief. Aur volume **double** ho gaya (10,633 → 21,307).

Insight 3 — Kis company ko complaint karte ho, ye zyada matter karta hai

Chi-square se prove kiya: company ka effect ($V = 0.341$) product se **1.8 guna strong** ($V = 0.186$). Relief rate **0.0% se 38.5%** tak jaata hai 112 companies mein.

Company	Monetary relief rate	Untimely rate
Bank of America	32.7%	1.83%
Citibank	26.8%	0.00%
American Express	25.5%	0.39%
Wells Fargo	9.9%	0.01%
Capital One	8.8%	0.01%

Bank of America Capital One se **3.7 guna** zyada rate pe monetary relief deta hai.

Insight 4 — Debt/credit management sabse tezi se badh raha hai

Product	2023	2025	Change
Debt or credit management	484	4,267	+781.6%
Student loan	10,633	21,307	+100.4%
Payday / personal loan	7,242	13,138	+81.4%
Credit card	54,409	89,976	+65.4%
Mortgage	22,853	24,697	+8.1%

Growth rate dekho, level nahi. Ye category abhi sirf 1.1% volume hai lekin 780% grow kar rahi hai — 2 saal mein material problem ban jayegi, aur har volume-ranked dashboard ise insignificant dikhata rahega jab tak late na ho jaye.

Honest limitations — ye bolna aapko credible banata hai

INTERVIEW MEIN YE ZAROOR MENTION KARNA

1. **Size-adjusted metric business-model sensitive hai.** Assets size proxy hai, customer count nahi. Agar customer data hota toh ranking compress ho jaati.
2. **Complaint volume complaint karne ki propensity bhi reflect karta hai** — educated, higher-income customers zyada complaint karte hain. Kuch company effect actually customer-mix effect ho sakta hai.
3. **50% volume ka koi FDIC denominator hai hi nahi.** MOHELA — sabse kharab servicer — headline metric mein by construction invisible hai.
4. **CFPB ne 2017 mein consumer-dispute flag band kar diya**, toh "consumer resolution se satisfied tha ya nahi" measure nahi ho sakta. Untimely aur no-relief substitutes hain, equivalent nahi.

Interview Q&A — 20 sawal aur jawab

Q1. Is project ke baare mein 2 minute mein batao

"Maine CFPB ke public complaint database se 6.24 lakh US banking complaints analyze kiye, 3 saal ka data. Core question ye tha: kaun si companies sabse zyada consumer risk generate karti hain — **size adjust karne ke baad.**"

"Pipeline aisa hai: CFPB API se data pull kiya cursor pagination se, pandas mein clean kiya, FDIC ke bank asset data se fuzzy matching karke enrich kiya, PostgreSQL mein COPY se load kiya, SQL window functions se analyze kiya, chi-square se validate kiya, aur Tableau dashboard banaya."

"Main finding ye thi ki **volume risk signal nahi hai, size proxy hai**. Synchrony Financial volume pe 7th hai lekin per dollar assets ke hisaab se desh mein pehle number pe — peer average se 4.6 guna. Aur JPMorgan volume pe 2nd worst dikhta hai lekin size-adjust karne pe 28 mein se 25th hai, matlab best mein se ek."

Q2. Aapne 87% data kyun hata diya? Kya wo cheating nahi hai?

"Ye scope decision tha, cherry-picking nahi — aur maine README mein poora document kiya hai."

"Hataye gaye 82 lakh complaints credit reporting ke the, jo Equifax, Experian, TransUnion ke against hain. Ye **banks nahi hain** — credit bureaus hain. Inke paas FDIC insured assets nahi hote, toh mera headline metric inke liye calculate hi nahi ho sakta."

"Agar main inhe rakhta toh do problems hoti. Ek — 87% volume hone ki wajah se ye har product, har state, har trend chart ko dabaa dete. Do — mera size-adjusted analysis inke liye undefined hota, toh main unhe silently drop karta ya galat rank karta."

"Maine README mein clearly likha hai ki ye ab **banking risk analysis hai, whole-of-CFPB analysis nahi**. Aur maine ye bhi likha ki mere volume numbers headline CFPB figures se match nahi karenge, kyunki wo credit reporting se drive hote hain."

Q3. Fuzzy matching mein accuracy kaise ensure ki?

"Teen tarah se. Pehla — maine **output review kiya**, sirf score pe bharosa nahi kiya. Isi se Ocwen/Owen ka bug pakda, jo 96.6 score pe match hua tha lekin galat tha."

"Doosra — maine guards **strict** rakhe. Exact normalized match, ya kam se kam 2 distinctive tokens plus Jaccard 0.6. Iska cost ye tha ki match rate kam ho gaya, lekin ye affordable tha kyunki exact matches aur curated overrides milke 99% volume carry karte hain."

"Teesra — ek **plausibility sweep** banaya jo 300 complaints per \$1B se zyada wali kisi bhi matched company ko flag karta hai. Koi real bank itna generate nahi kar sakta. Abhi ye khaali return karta hai, aur main isko report karta hoon auto-correct nahi — taaki regression dikhe."

Q4. Chi-square kyun use kiya aur p-value kya tha?

"Chi-square test of independence use kiya ye check karne ke liye ki complaint outcome company se independent hai ya nahi. Lekin — aur ye important hai — **maine p-value pe conclusion nahi banaya.**"

"6 lakh sample pe chi-square kisi bhi association pe p less than 0.001 dega, chahe wo kitni bhi choti ho. Significance guaranteed hai, isliye wo koi information nahi deta. Mera p-value literally 1e-300 se kam tha."

"Isliye maine **Cramér's V** report kiya — effect size. Company ke liye 0.341 (strong), product ke liye 0.186 (weak). Ratio 1.8. Ye business statement banata hai: aap kis company ko complaint karte ho ye product se 1.8 guna zyada matter karta hai."

"Aur maine standardized residuals bhi nikale, kyunki chi-square poore table ka ek number deta hai — ye nahi batata ki problem kahan hai. Residuals se dikha ki mortgage (−66) aur student loan (−62) expected se bahut kam relief dete hain."

Q5. 6 lakh rows Postgres mein kaise load kiye? Kitna time laga?

"`copy_expert` use kiya psycopg2 mein — PostgreSQL ka COPY command. 17 seconds mein 6.24 lakh rows, matlab **36,700 rows per second.**"

"Maine pandas ka `to_sql` deliberately avoid kiya. Wo batched INSERTs karta hai jo is size pe minutes leta hai aur WAL flood karta hai. COPY poori CSV ko ek single statement mein stream kar deta hai — file Python mein parse hoti hi nahi."

"Load ke baad 9 indexes banaye, jisme do composite hain jo actual query shapes se match karte hain, aur ANALYZE chalaya taaki planner ke statistics fresh rahein."

Q6. Missing values kaise handle kiye?

"Sabse pehle — maine **missing values ki wajah se ek bhi row drop nahi ki.**"

"Kyunki is dataset mein 'missing' zyadatar ek real state encode karta hai, lost data nahi. Sabse acha example `tags` column hai — 83% null. Lekin us field mein sirf 'Servicemember' ya 'Older American' aata hai. Null ka matlab hai 'koi bhi apply nahi hota'. Agar main mode se impute karta toh 5 lakh **fake veterans** ban jaate. Aur drop karta toh 83% dataset chala jaata."

"Toh maine har column ka alag decision liya aur reasoning ke saath log kiya. tags ko 'No tag' se fill kiya plus do boolean flags banaye. `company_public_response` ko 'No public response' se — kyunki company ka chup rehna khud ek information hai."

Q7. Duplicates kaise define kiye?

"`complaint_id` — source ki apni primary key — pe zero duplicates the. Toh maine business key define ki: full timestamp, company, product, sub-product, issue, sub-issue, state, ZIP, aur submission channel."

"Is key pe 19 rows match hui — same company, same issue, **same second**. Do alag log ka 9 fields pe same second mein match hona plausible nahi hai, toh ye double submissions hain. Hata diye."

"Lekin agar main timestamp ko sirf date tak truncate karta toh **9,568 rows** duplicate dikhti. Maine wo **jaan-boojh ke nahi hataye** — kyunki wo alag log hain jo same din same bade bank ke against complaint kar rahe hain, jo bilkul expected hai."

"Ye important hai kyunki unhe hataana **directional bias** create karta: sabse zyada complaints wali companies se sabse zyada rows delete hoti, matlab jo companies sabse kharab hain unki ranking automatically improve ho jaati."

Q8. Agar aapko ye project dobara karna hota toh kya alag karte?

"Teen cheezein."

"Ek — **matching guards pehle se plan karta**. Maine 5 guards banaye lekin sab reactive the, har ek kisi false positive ke baad. Agar main pehle se soch leta ki fuzzy matching ke failure modes kya hain toh time bachta."

"Do — **customer count data dhundhta**. Assets size ka acha proxy hai lekin perfect nahi. Agar deposit accounts ya cardholder numbers milte toh metric aur clean hota. Maine ye limitation document ki hai."

"Teen — **automated tests likhta**. Abhi mere paas acceptance checks hain har script ke end mein, jo kaafi ache hain, lekin proper pytest suite hoti toh regression pakadna aur aasan hota."

Q9. Sabse mushkil technical challenge?

"Entity resolution — CFPB companies ko FDIC institutions se match karna."

"Problem structural thi: CFPB holding company ka naam deta hai ('U.S. BANCORP'), FDIC insured subsidiary ka ('U.S. Bank National Association'). Ye same entity hain lekin naam alag."

"Maine FDIC ka `NAMEHCR` field use kiya jo holding company ka naam carry karta hai. Lekin usme abbreviations hain — 'U S BCORP' matlab 'U.S. Bancorp'. Toh pehle abbreviations expand ki, phir rapidfuzz se match kiya."

"Aur ek trap tha: **683 institutions ke paas koi holding company hai hi nahi**, unka field blank hai. Agar blank pe group by karta toh 683 alag banks ek fake \$822 billion entity ban jaate. Maine unhe unke apne CERT number se key kiya."

Q10. Aapke insights actionable kaise hain?

"Har insight ke saath maine ek recommendation di hai jo **specific audience** ke liye hai, generic nahi."

"Jaise pehle insight ke liye: 'risk teams complaint counts pe ranking band karein, complaints per \$1B ko primary screening metric banayein, aur jin institutions ka size-adjusted rank volume rank se 10+ upar hai unko review karein.' Ye specific hai — SoFi, Barclays, Santander naam ke saath."

"Student loan insight ke liye: 'response timeliness ko complaint volume se independent supervisory trigger banao. MOHELA akele 4,632 late responses ka hissa hai.'"

"Aur maine ek section likha hai '**What would change my conclusions**' — 5 limitations jo mere analysis ko weak kar sakti hain. Ye isliye kiya kyunki agar koi reviewer unstated limitation pakadta hai toh wo sochta hai maine miss ki. Agar main khud batata hoon toh wo competence dikhta hai."

Q11. Data quality kaise verify ki?

"Har stage ke end mein acceptance checks likhe jo pass/fail print karte hain."

"Stage 1 mein 13 checks: row count API total se exact match, zero duplicate IDs, zero unparsed dates, har product ka count scope table se match, har narrative ID ka parent complaint exist karta hai."

"Stage 2 mein 9 checks. Ek check ne actually bug pakda — 'nulls sirf zip3 mein hone chahiye' fail ho gaya, aur investigate karne pe pata chala ki maine column ko `'None'` string se fill kiya tha, jo pandas ke default NA values mein hai. Matlab CSV likh ke padho toh nulls wapas aa jaate the — fill khud ko undo kar raha tha."

"Stage 4 mein 7 database-level checks, jisme information_schema se verify karta hoon ki dates actually DATE type hain aur booleans BOOLEAN."

Q12. Risk score ka formula kya hai aur weights kaise chune?

"Score percentile-based hai — matlab 80 ka score ka matlab '80% peers se kharab'. Percentile isliye use kiya kyunki ye rates heavily right-skewed hain; ek company 100% untimely pe hai, toh min-max scale mein baaki sab neeche da6 jaate."

"Weights: no-relief 40%, untimely 35%, disputes-facts 15%, routing days 10%. Maine ye **explicitly document kiye hain ek judgement ke taur pe, derivation ke taur pe nahi** — taaki koi reader weighting se disagree kar sake bajaye reverse-engineer karne ke. Underlying rates sab output mein hain, toh koi bhi re-weight kar sakta hai."

"Aur ek component maine **test karke hataya**. explanation_only_rate 20% pe tha, lekin wo no_relief se 0.954 correlate kar raha tha — ek hi cheez do baar. Dono milke 55% weight le rahe the aur untimely ko crowd out kar rahe the. Hataane ke baad ranking sahi hui."

Q13. Do scores kyun banaye, ek kyun nahi?

"Kyunki ek single composite do tarah se misleading hota."

"Pehla — size-adjusted metric assets se divide karta hai, jo balance sheet size hai customer count nahi. Ek monoline credit card company lakhs customers serve karti hai chhote balance sheet pe. Ek universal bank ke paas mortgages hain jo assets badhate hain bina retail customers badhaye. Toh card issuers **business model ki wajah se** upar aate hain."

"Doosra — wo metric 50% complaint volume ke liye undefined hai, kyunki non-banks ke FDIC assets hote hi nahi. Single composite chupchap aadha market drop kar deta."

"Isliye **conduct_risk_score** banaya jo sirf resolution behaviour dekhta hai aur sabke liye valid hai — yahi main conduct claims ke liye use karta hoon. Aur **exposure_risk_score** jo size bhi add karta hai, sirf matched banks ke liye."

Q14. Tableau mein business logic kyun nahi rakha?

"Kyunki Tableau calculated fields versioned nahi hote aur review karna mushkil hai."

"Maine 4 PostgreSQL views banaye. 500-complaint floor, resolved-only denominator, trend window bound, FDIC join — sab SQL mein hain. Agar 'risk' ki definition badalti hai toh view badlo, saare worksheets automatically follow karenge."

"Agar ye logic Tableau mein hota toh kisi ko yaad rakhna padta ki kaun se calculated field mein kaun sa rule chhupa hai — aur ek worksheet update karna bhool jaate toh dashboard inconsistent ho jaata."

Q15. Pipeline reproducible kaise hai?

"Chaar tarah se."

"Ek — **numbered scripts**, har ek idempotent aur re-runnable. Script 01 checkpointed hai, toh interrupted download resume hota hai."

"Do — **interpreter aur dependencies pinned**. requirements-lock.txt verified install se generate hui hai."

"Teen — **logs script se generate hote hain**. cleaning_log.md ke numbers 02_clean.py chalane pe likhe jaate hain. Agar data badla toh script chalo, numbers apne aap update. Ye kabhi stale nahi ho sakte."

"Chaar — **secrets .env mein hain** jo gitignored hai, aur .env.example committed hai taaki koi aur setup kar sake."

Q16. Jan 2025 spike ka kya kiya?

"Pehle investigate kiya. 18,441 extra complaints the, aur **67% sirf do companies se** — Navy Federal 484 se 7,725 (16 guna) aur Capital One 840 se 5,876. 15 se 18 January mein concentrated, 17 tarikh ko akele 4,978. Dominant issue overdraft/NSF fees."

"Ye signature hai **coordinated filing campaign** ki — aam taur pe kisi enforcement action ke baad hoti hai, service quality degrade hone ka signal nahi."

"Maine data **delete nahi kiya** kyunki complaints real hain. Balki flag kiya, aur sensitivity check chalaya — mahina hata ke ranking dobara nikali. **28 mein se 28 companies ka rank same raha**, toh ranking robust hai."

"Aur maine Tableau dashboard pe annotation add karne ka instruction likha hai, kyunki bina uske koi bhi viewer us spike ko real service collapse samjhega."

Q17. Sabse bada business insight kya hai?

"Ki **volume aur risk alag cheezein hain, aur poori industry volume dekh rahi hai**."

"Concrete example: Wells Fargo aur JPMorgan ko lagbhag identical complaints mile — 41,051 aur 40,391. Koi bhi volume-based dashboard inhe same risk dikhayega. Lekin JPMorgan ke paas 2.2 guna assets hain, toh per dollar Wells Fargo 2.2 guna friction generate karta hai."

"Aur ulta bhi sach hai — Synchrony volume pe 7th hai, koi notice nahi karega, lekin per dollar wo **desh mein number one** hai, peer average se 4.6 guna."

"Iska practical matlab ye hai ki agar aap complaint counts se monitor kar rahe ho toh aap systematically chhote lekin riskier institutions ko miss kar rahe ho."

Q18. Aapne kaun se SQL concepts use kiye?

"Window functions kaafi use kiye — `RANK()` , `DENSE_RANK()` ties dikhane ke liye, `NTILE(4)` quartiles ke liye, `ROW_NUMBER() OVER (PARTITION BY ...)` har group ka top-N nikalne ke liye."

"Trend analysis ke liye `LAG(1)` previous month, `LAG(12)` year-over-year, `LEAD()` , aur moving average ke liye `AVG() OVER (ORDER BY ... ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)` ."

" `FILTER (WHERE ...)` aggregates kaafi use kiye — ye `CASE WHEN` se cleaner hai conditional counts ke liye."

"Aur ek cheez jo maine seekhi: `PERCENTILE_CONT` ek **ordered-set aggregate** hai, PostgreSQL use window function ki tarah allow nahi karta. Toh peer median ko alag CTE mein nikal ke cross join karna pada."

Q19. Ye project real world mein kaise use hoga?

"Teen tarah se."

"**Regulator ke liye:** supervisory prioritization. Abhi agar CFPB volume dekh ke supervision allocate karta hai toh bade banks pe focus jaata hai. Size-adjusted metric se pata chalta hai ki kaun se institutions apne size ke hisaab se outlier hain."

"**Bank ki compliance team ke liye:** peer benchmarking. Agar aapka relief rate peers se bahut kam hai toh wo ek defensible-conduct question hai jo kisi volume dashboard mein nahi dikhega."

"**Consumer advocacy ke liye:** transparency. Ek consumer ko nahi pata ki uske remediation ke chances issuer ke hisaab se 4 guna vary karte hain."

Q20. Aapne is project se kya seekha?

"Sabse badi seekh ye hai ki **data analysis mein sabse khatarnak galtiyan wo hain jo galat nahi dikhti.**"

"Product rename wala case perfect example hai. Chart bilkul normal dikh raha tha. Ek clean line jo badh rahi thi. Agar main sirf chart dekhta toh confidently likh deta ki credit card complaints mein 300% surge aaya. Koi error message nahi, koi red flag nahi — bas ek galat conclusion."

"Isi tarah Ocwen/Owen match. 96.6 score tha — algorithm confident tha. Sirf ye ki number absurd tha, isliye maine check kiya."

"Toh maine ye habit banayi ki har stage ke baad output **manually review karo** aur poochho 'kya ye number believable hai?'. Aur automated checks likho jo assumptions verify karein. Mere har script ke end mein acceptance checks hain, aur unme se do ne actual bugs pakde."

Resume bullets aur closing

Option A — analytical result pe focus

Built an end-to-end pipeline analysing **624,708 CFPB banking complaints** (Python, PostgreSQL, Tableau), enriching them with FDIC asset data via fuzzy entity matching to compute a size-adjusted risk metric; showed that **raw complaint volume misranks the industry** — the 2nd-largest complaint generator ranks **25th of 28 once normalised by assets**, while the worst performer runs at **4.6× the peer average** — and proved via chi-square (Cramér's V 0.34 vs 0.19, n=624k) that complaint outcome depends **1.8× more on the company than the product**.

Option B — engineering pe focus

Engineered a reproducible 7-stage data pipeline ingesting **624,727 records** from the CFPB API (cursor pagination, retry/backoff, checkpointed resume), cleaned and validated them into a typed **PostgreSQL** warehouse via bulk `COPY` at **~36,700 rows/sec** with 9 indexes and automated integrity checks; built entity resolution against **4,255 FDIC institutions** with rapidfuzz, and delivered SQL views powering a live-connected **Tableau** dashboard plus a statistical notebook.

Skills jo ye project demonstrate karta hai

Category	Skills
Data Engineering	REST API integration, cursor pagination, retry/backoff, checkpointing, ETL design, bulk loading (COPY), indexing strategy, schema design
Data Cleaning	Missing value strategy, deduplication logic, taxonomy normalization, data quality flags, entity resolution
SQL	Window functions, CTEs, FILTER aggregates, percentile functions, views, query optimization
Statistics	Chi-square test, effect size (Cramér's V), standardized residuals, Spearman correlation, sensitivity analysis
Business Analysis	Metric design, normalization, actionable recommendations, limitation disclosure
Tools	Python, pandas, PostgreSQL, Tableau, Git, Jupyter

AAKHRI ADVICE — INTERVIEW KE LIYE

1. Agar ek hi cheez yaad rakhni ho toh ye: *"Wells Fargo aur JPMorgan ko same complaints mile, lekin JPMorgan ke paas 2.2 guna assets hain — toh per dollar Wells Fargo 2.2 guna zyada risk hai."* Ye ek line poora project explain kar deti hai.
2. Jab bhi mauka mile, **bugs ke baare mein baat karo**. "Maine pipeline banaya" common hai. "Maine apne hi code mein ye galti pakdi aur fix ki" rare hai.
3. Limitations khud batao. Ye weakness nahi, **credibility** hai. Jo analyst apni analysis ki seemayein jaanta hai, wo zyada trustworthy hota hai.
4. Numbers yaad rakho: 6.24 lakh rows, 3,193 companies, 36 months, 147.8 per \$1B (Synchrony), 0.34 vs 0.19 (Cramér's V), 36,700 rows/sec (load speed).

All the best, Sahil! 

github.com/sahilsharma0309/banking-complaints-analysis

Ye project sirf ek dashboard nahi hai — ye ek complete analytical argument hai, jo data se shuru hoke actionable recommendation tak jaata hai.